Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

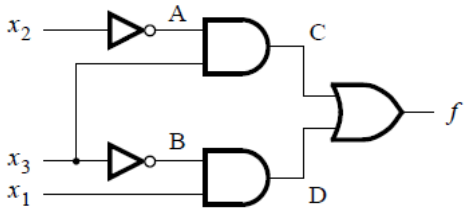
Problem Set 1

September 6, 2022

1. Use algebraic manipulation to prove that $x + yz = (x + y)(x + z)$. This is the distributive rule, identity 12b in Section 2.5 of textbook.

2. Use the Venn diagram to prove DeMorgan's theorem, identity 15a in Section 2.5 of textbook.

3. Draw a timing diagram for the following circuit. Show the waveforms that can be observed on all wires in the circuit.



4. Use algebraic manipulation to show that for three input variables x_1, x_2, and x_3:

$$\sum m(1,2,3,4,5,6,7) = x_1 + x_2 + x_3$$

5. Use algebraic manipulation to find the minimum product-of-sums expression for the function:

$$f = (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + x_3)(\bar{x}_1 + \bar{x}_2 + x_3)(x_1 + x_2 + \bar{x}_3)$$

6. Show the location of all minterms in a three-variable Venn diagram.

7. Design the simplest product-of-sums expression for the function:

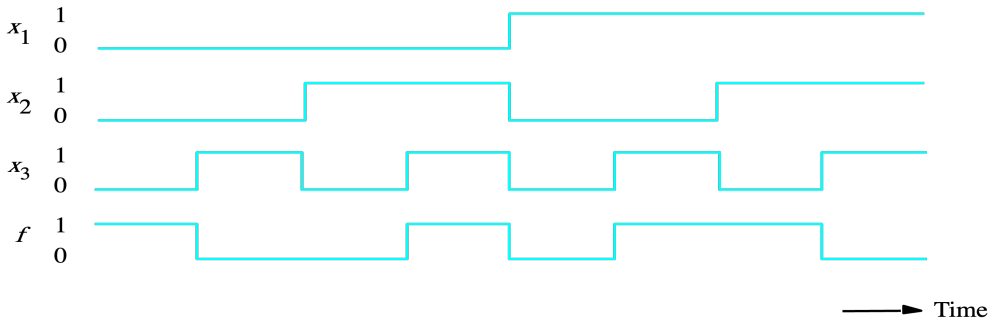
$$f(x_1, x_2, x_3) = \prod M(0,1,5,7)$$

8. Derive the simplest sum-of-products expression for the function:

$$f(x_1, x_2, x_3, x_4) = x_1\bar{x}_3\bar{x}_4 + x_2\bar{x}_3x_4 + x_1\bar{x}_2\bar{x}_3$$

9. Design the simplest circuit that has three inputs, x_1, x_2, and x_3, which produces an output value of 1 whenever exactly one or two of the input variables have the value 1; otherwise, the output has to be 0.

10. For the following timing diagram, synthesize the function $f(x_1, x_2, x_3)$ in the simplest sum-of-products form.



11. Design a circuit with output f and inputs x_1, x_0, y_1, and y_0. Let $X = x_1x_0$ be a number where the four possible values of X, namely, 00, 01, 10, and 11, represent the four numbers, 0, 1, 2, and 3, respectively. Similarly, $Y = y_1y_0$ represents another number with the same four possible values. The output f should be 1 if the numbers represented by X and Y are equal. Otherwise, f should be 0.

    (a) Show the truth table for f.
    (b) Synthesize the simplest possible product-of-sums expression for f.

12. Implement the function for the following truth table using only NAND gates.

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

13. Write the VHDL code to implement the function:

$$f(x_1, x_2, x_3) = \sum m(0,1,3,4,5,6).$$

1) Prove that $x + yz = (x+y)(x+z)$

RHS: $(x+y)(x+z) = xx + xz + xy + yz$ — 12a Distributive

$= x + xz + xy + yz$ — 7a Absorption (Axiomatic)

$= x(1 + z + y) + yz$ — 12a Distributive

$= x + yz$ — 5b & 6a Absorption (Axiomatic)

$\therefore$ LHS = RHS $\square$

1a.	$0 \cdot 0 = 0$	1b.	$1 + 1 = 1$
2a.	$1 \cdot 1 = 1$	2b.	$0 + 0 = 0$
3a.	$0 \cdot 1 = 1 \cdot 0 = 0$	3b.	$1 + 0 = 0 + 1 = 1$
4a.	If $x = 0$ then $\overline{x} = 1$	4b.	If $x = 1$ then $\overline{x} = 0$

5a.	$x \cdot 0 = 0$	5b.	$x + 1 = 1$
6a.	$x \cdot 1 = x$	6b.	$x + 0 = x$
7a.	$x \cdot x = x$	7b.	$x + x = x$
8a.	$x \cdot \overline{x} = 0$	8b.	$x + \overline{x} = 1$
9.	$\overline{\overline{x}} = x$		

10a.	$x \cdot y = y \cdot x$	10b.	$x + y = y + x$ — *Commutative*
11a.	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	11b.	$x + (y + z) = (x + y) + z$ — *Associative*
12a.	$x \cdot (y + z) = x \cdot y + x \cdot z$	12b.	$x + y \cdot z = (x + y) \cdot (x + z)$ — *Distributive*
13a.	$x + x \cdot y = x$	13b.	$x \cdot (x + y) = x$ — *Absorption*
14a.	$x \cdot y + x \cdot \overline{y} = x$	14b.	$(x + y) \cdot (x + \overline{y}) = x$ — *Combining*
15a.	$\overline{x \cdot y} = \overline{x} + \overline{y}$	15b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$ — *DeMorgan's Thm*
16a.	$x + \overline{x} \cdot y = x + y$	16b.	$x \cdot (\overline{x} + y) = x \cdot y$
17a.	$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$	17b.	$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$ — *Consensus*

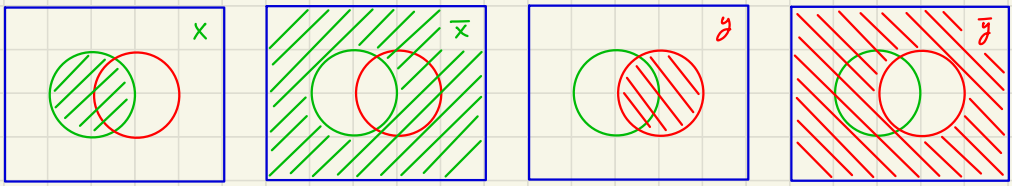
$\cup \equiv \vee$ OR $\cap \equiv \wedge$ AND

2) **DeMorgan's Theorem**: Describes the interchange b/w the union & intersection of sets under their complements

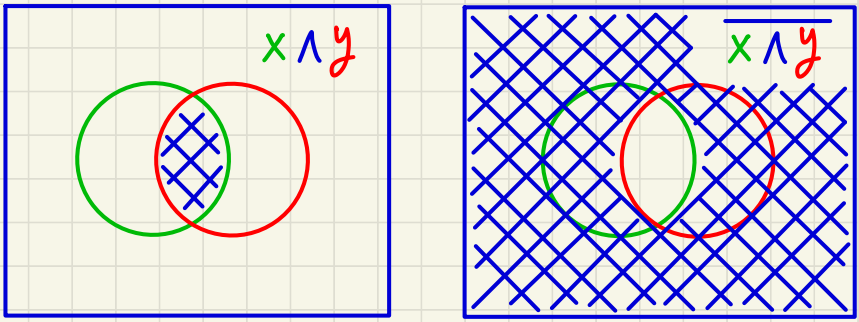
Basically, $\overline{x \wedge y} \equiv \bar{x} \vee \bar{y}$; $\overline{x \vee y} \equiv \bar{x} \wedge \bar{y}$ Super important for optimization

Prove Eq 15a, $\overline{x \wedge y} = \bar{x} \vee \bar{y}$, w/ Venn diagrams

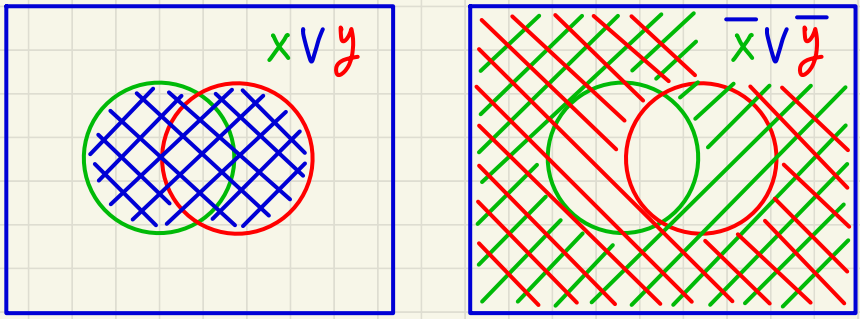
let $x \rightarrow$ ◯ & $y \rightarrow$ ◯



LHS:

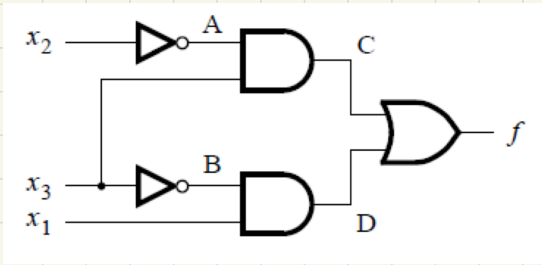


RHS:

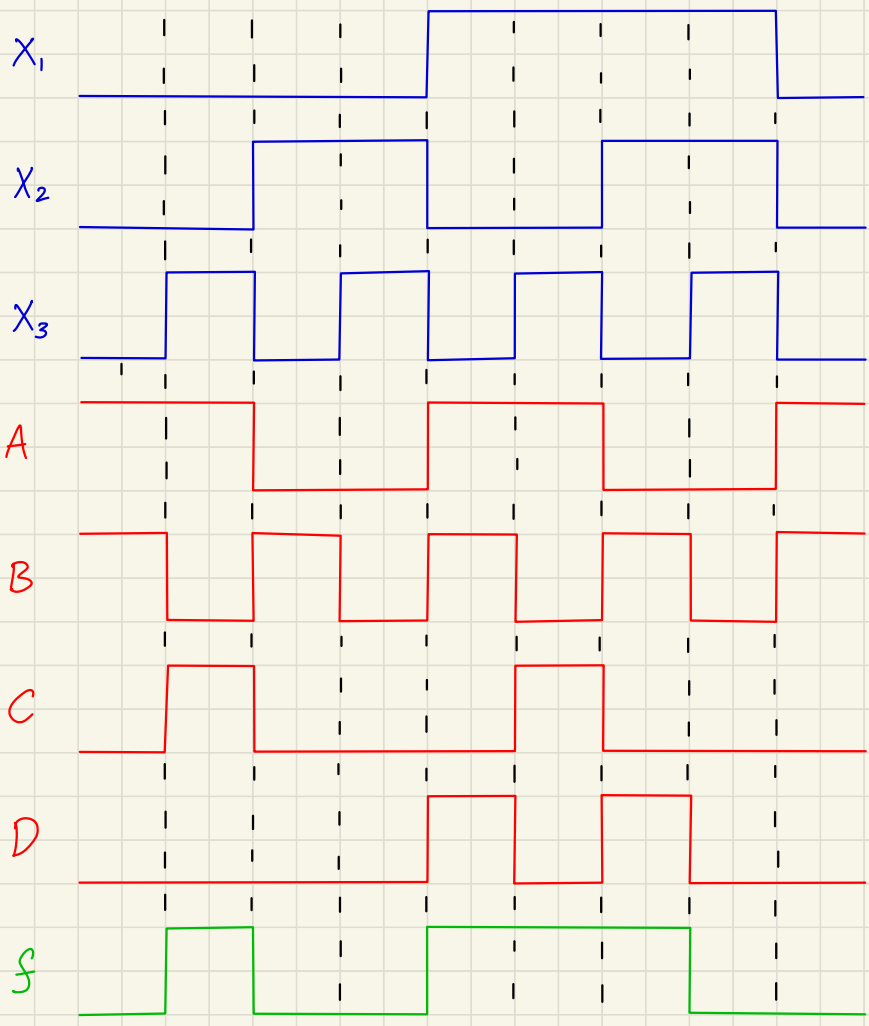


$\therefore$ LHS = RHS □

3) Draw a timing diagram



X_1	X_2	X_3	A	B	C	D	f
ø	ø	ø	1	1	ø	ø	ø
ø	ø	1	1	ø	1	ø	1
ø	1	ø	ø	1	ø	ø	ø
ø	1	1	ø	ø	ø	ø	ø
1	ø	ø	1	1	ø	1	1
1	ø	1	1	ø	1	ø	1
1	1	ø	ø	1	ø	1	1
1	1	1	ø	ø	ø	ø	ø



4) Show that $\sum m(1,2,3,4,5,6,7) = x_1 + x_2 + x_3$ w/ algebra SoP

$$\sum m(1,2,3,4,5,6,7) = m_1 + m_2 + m_3 + m_4 + m_5 + m_6 + m_7$$

$$= \underbrace{\overline{x}_1 \overline{x}_2 x_3}_{001=1} + \underbrace{\overline{x}_1 x_2 \overline{x}_3}_{010=2} + \underbrace{\overline{x}_1 x_2 x_3}_{011=3} + \underbrace{x_1 \overline{x}_2 \overline{x}_3}_{100=4} +$$

$$+ \underbrace{x_1 \overline{x}_2 x_3}_{101=5} + \underbrace{x_1 x_2 \overline{x}_3}_{110=6} + \underbrace{x_1 x_2 x_3}_{111=7}$$

$$= x_1(\overline{x}_2\overline{x}_3 + \overline{x}_2 x_3 + x_2\overline{x}_3 + x_2 x_3) + x_2(\overline{x}_1\overline{x}_3 + \overline{x}_1 x_3 + x_1\overline{x}_3 + x_1 x_3) +$$

$$+ x_3(\overline{x}_1\overline{x}_2 + \overline{x}_1 x_2 + x_1\overline{x}_2 + x_1 x_2)$$

Use the axiomatic defs; $x + \overline{x} = 1$, $x\overline{x} = \emptyset$, $xx = x$

& $z(\overline{x}\overline{y} + \overline{x}y + x\overline{y} + xy) = z(\overline{x}(\overline{y} + y) + x(\overline{y} + y)) = z$

$$= x_1 + x_2 + x_3$$

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2 x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1 x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1 x_2 x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2 x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1 x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

DeMorgans $\overline{m} = M$

5) Simplify $f = (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(\overline{x}_1 + \overline{x}_2 + x_3)(x_1 + x_2 + \overline{x}_3)$ w/ algebra

$$f = (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(\overline{x}_1 + \overline{x}_2 + x_3)(x_1 + x_2 + \overline{x}_3)$$

Re-order

$$= [(x_1 + x_2) + x_3] \cdot [(x_1 + \overline{x}_2) + \overline{x}_3] \cdot [(\overline{x}_2 + x_3) + x_1][(\overline{x}_2 + x_3) + \overline{x}_1]$$

Distribute

$$= [\underbrace{(x_1 + x_2) + (x_1 + x_2)}_{x + x = x}(x_3 + \overline{x}_3)^{1} + x_3\overline{x}_3^{\,0}] \cdot [x_1\overline{x}_1^{\,0} + \underbrace{(\overline{x}_2 + x_3)(x_1 + \overline{x}_1)^{1} + (\overline{x}_2 + x_3)}_{x + x = x}]$$

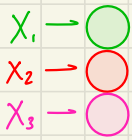
Axiomatic

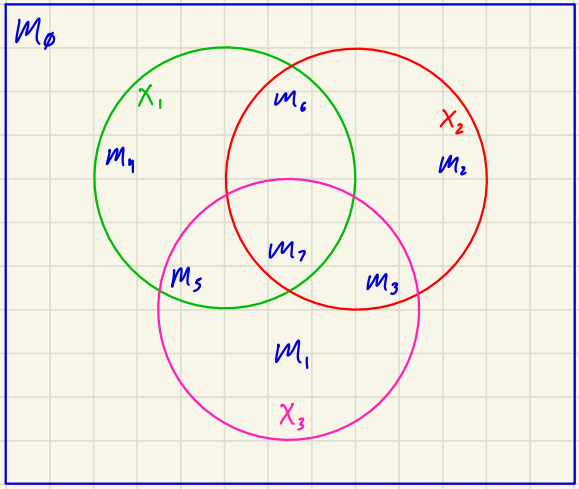
$$= (x_1 + x_2)(\overline{x}_2 + x_3)$$

6) Let variables be X_1, X_2, X_3 ; for 3 variables we have 3 bits ∴ we have 8 min terms, m_i $\forall i \in [\emptyset, 7] \cap \mathbb{Z}$. To show the locations we can look at the truth table.

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2 x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1 x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1 x_2 x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2 x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1 x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

So if the variable has a 1 then the m_i is included in its set in the Venn diagram





	X_1	X_2	X_3
m_0	ø	ø	ø
m_1	ø	ø	1
m_2	ø	1	ø
m_3	ø	1	1
m_4	1	ø	ø
m_5	1	ø	1
m_6	1	1	ø
m_7	1	1	1

7) Simplify $f(x_1, x_2, x_3) = \prod M(0,1,5,7)$ w/ algebra PoS

$\prod M(0,1,5,7) = M_0 \cdot M_1 \cdot M_5 \cdot M_7$

$= (X_1 + X_2 + X_3) \cdot (X_1 + X_2 + \overline{X}_3) \cdot (\overline{X}_1 + X_2 + \overline{X}_3) \cdot (\overline{X}_1 + \overline{X}_2 + \overline{X}_3)$

$111 = 000 = 0 \qquad 110 = 001 = 1 \qquad \overline{010} = 101 = 5 \qquad \overline{000} = 111 = 7$

Re-order

$= [(X_1 + X_2) + X_3] \cdot [(X_1 + X_2) + \overline{X}_3] \; [(\overline{X}_1 + X_2) + \overline{X}_3] \cdot [(\overline{X}_1 + \overline{X}_2) + \overline{X}_3]$

Distribute

$= [(X_1 + X_2)(X_1 + X_2) + X_3\overline{X}_3 + X_3(X_1 + X_2) + \overline{X}_3(X_1 + X_2)]$

$xx = x$

$\cdot [(\overline{X}_1 + X_2)(\overline{X}_1 + \overline{X}_2) + \overline{X}_3(1 + \overline{X}_1 + X_2 + \overline{X}_1 + \overline{X}_2)]$

$= [(X_1 + X_2)(1 + X_3 + \overline{X}_3)] \cdot [\overline{X}_1 + \overline{X}_1\overline{X}_2 + X_2\overline{X}_2 + \overline{X}_3]$

Rule 8a: $X + XY = X(1 + Y) = X$

$= (X_1 + X_2)(\overline{X}_1 + \overline{X}_3)$

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

10a.	$x \cdot y = y \cdot x$	10b.	$x + y = y + x$	Commutative
11a.	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	11b.	$x + (y + z) = (x + y) + z$	Associative
12a.	$x \cdot (y + z) = x \cdot y + x \cdot z$	12b.	$x + y \cdot z = (x + y) \cdot (x + z)$	Distributive
13a.	$x + x \cdot y = x$	13b.	$x \cdot (x + y) = x$	Absorption
14a.	$x \cdot y + x \cdot \overline{y} = x$	14b.	$(x + y) \cdot (x + \overline{y}) = x$	Combining
15a.	$\overline{x \cdot y} = \overline{x} + \overline{y}$	15b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$	DeMorgan's Thm
16a.	$x + \overline{x} \cdot y = x + y$	16b.	$x \cdot (\overline{x} + y) = x \cdot y$	
17a.	$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$	17b.	$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$	Consensus

8) Simplify $f(x_1, x_2, x_3, x_4) = x_1\overline{x}_3\overline{x}_4 + x_2\overline{x}_3x_4 + x_1\overline{x}_2\overline{x}_3$

$$f = x_1\overline{x}_3\overline{x}_4 + x_2\overline{x}_3x_4 + x_1\overline{x}_2\overline{x}_3 = x_1\overline{x}_3\underbrace{(\overline{x}_4 + \overline{x}_2)}_{\text{DeMorgans}} + x_2\overline{x}_3x_4$$

$$= x_1\overline{x}_3 \cdot \overline{x_2x_4} + x_2\overline{x}_3x_4 = \overline{x}_3\underbrace{(x_1 \cdot \overline{x_2x_4} + x_2x_4)}_{\text{Rule 16a}}$$

$$= \overline{x}_3(x_1 + x_2x_4) = x_1\overline{x}_3 + x_2\overline{x}_3x_4$$

10a.	$x \cdot y = y \cdot x$	10b.	$x + y = y + x$ — *Commutative*
11a.	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	11b.	$x + (y + z) = (x + y) + z$ — *Associative*
12a.	$x \cdot (y + z) = x \cdot y + x \cdot z$	12b.	$x + y \cdot z = (x + y) \cdot (x + z)$ — *Distributive*
13a.	$x + x \cdot y = x$	13b.	$x \cdot (x + y) = x$ — *Absorption*
14a.	$x \cdot y + x \cdot \overline{y} = x$	14b.	$(x + y) \cdot (x + \overline{y}) = x$ — *Combining*
15a.	$\overline{x \cdot y} = \overline{x} + \overline{y}$	15b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$ — *DeMorgan's Thm*
16a.	$x + \overline{x} \cdot y = x + y$	16b.	$x \cdot (\overline{x} + y) = x \cdot y$
17a.	$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z$	17b.	$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z)$ — *Consensus*

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

9) Design a 3 input (X_1, X_2, X_3) circuit w/ output 1 if one or two of the input variables have a value of 1; else the output is Ø

Start w/ a truth table & then use PoS or SoP. Here if we did SoP we would have $\sum m(1,2,3,4,5,6)$, but since $\overline{m} = M$ then for PoS we would have $\prod M(0,7)$ ∴ lets go w/ PoS.

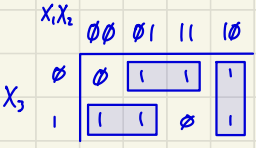
M_i	X_1	X_2	X_3	f
M_0	Ø	Ø	Ø	Ø
M_1	Ø	Ø	1	1
M_2	Ø	1	Ø	1
M_3	Ø	1	1	1
M_4	1	Ø	Ø	1
M_5	1	Ø	1	1
M_6	1	1	Ø	1
M_7	1	1	1	Ø

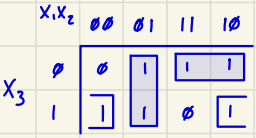
$$f = M_0 \cdot M_7 = (X_1 + X_2 + X_3)\cdot(\overline{X}_1 + \overline{X}_2 + \overline{X}_3)$$

$$= X_1(\overline{X}_1 + \overline{X}_2 + \overline{X}_3) + X_2(\overline{X}_1 + \overline{X}_2 + \overline{X}_3) + X_3(\overline{X}_1 + \overline{X}_2 + \overline{X}_3)$$

$$= X_1\overline{X}_2 + X_1\overline{X}_3 + \overline{X}_1X_2 + X_2\overline{X}_3 + \overline{X}_1X_3 + \overline{X}_2X_3$$

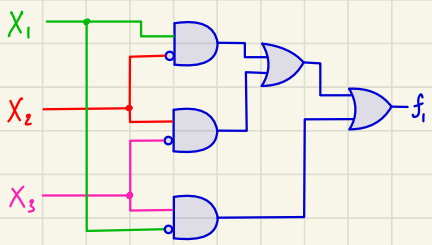
$$= \begin{cases} X_1\overline{X}_2 + X_2\overline{X}_3 + \overline{X}_1X_3 \rightarrow f_1 & \text{both are valid} \\ X_1\overline{X}_3 + \overline{X}_1X_2 + \overline{X}_2X_3 \rightarrow f_2 \end{cases}$$

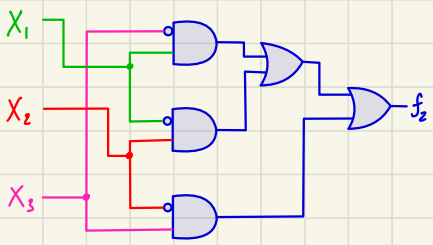




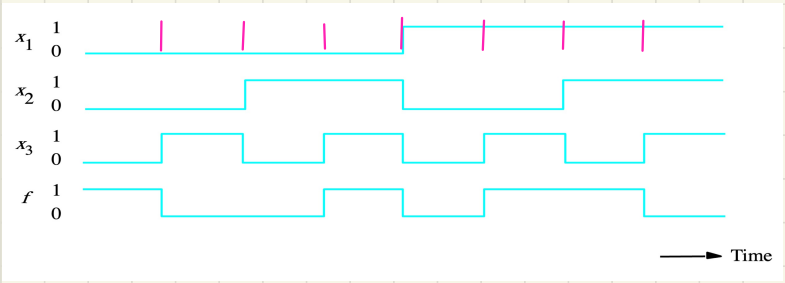
We can do this b/c some terms are redundant, so we only keep the Essential Prime Implicant, which are prime implicants that can only be covered by that prime implicant

Once we have the equation we can make the circuits pretty easily





10) Get an equation for the timing diagram



m_i	X_1	X_2	X_3	f
m_0	∅	∅	∅	1
m_1	∅	∅	1	∅
m_2	∅	1	∅	∅
m_3	∅	1	1	1
m_4	1	∅	∅	∅
m_5	1	∅	1	1
m_6	1	1	∅	1
m_7	1	1	1	∅

$$f = \sum m(\emptyset, 3, 5, 6) = m_\emptyset + m_3 + m_5 + m_6$$

$$= \overline{X}_1 \overline{X}_2 \overline{X}_3 + \overline{X}_1 X_2 X_3 + X_1 \overline{X}_2 X_3 + X_1 X_2 \overline{X}_3$$

$$= \overline{X}_1 + X_1 (\overline{X}_2 X_3 + X_2 \overline{X}_3)$$

11) Make a circuit with inputs x_0, x_1, y_0, y_1, where if we say $X = x_0 x_1$ & $Y = y_0 y_1$ then $f = 1$ when $X = Y$, otherwise $f = \emptyset$

11a) Make a truth table

x_1	x_0	y_1	y_0	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

11b) Derive the simplest P.S possible for f

$$f = \prod M(1,2,3,4,6,7,8,9,11,12,13,14)$$

This looks complicated if we multiply terms that have reductions like $X + \bar{X} = 1$ & $X\bar{X} = \emptyset$
then the problem b/m's much more manageable

$$M_2 M_3 = (x_1 + x_0 + \bar{y}_1 + y_0)(x_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (x_1 + x_0 + \bar{y}_1)$$
$$M_7 M_6 = (x_1 + \bar{x}_0 + \bar{y}_1 + \bar{y}_0)(x_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (x_1 + \bar{x}_0 + \bar{y}_1)$$
$$M_{12} M_{13} = (\bar{x}_1 + \bar{x}_0 + y_1 + y_0)(\bar{x}_1 + \bar{x}_0 + y_1 + \bar{y}_0) = (\bar{x}_1 + \bar{x}_0 + y_1)$$
$$M_8 M_9 = (\bar{x}_1 + x_0 + y_1 + y_0)(\bar{x}_1 + x_0 + y_1 + \bar{y}_0) = (\bar{x}_1 + x_0 + y_1)$$

$$f = M_1 (x_1 + x_0 + \bar{y}_1) M_4 (x_1 + \bar{x}_0 + \bar{y}_1)(\bar{x}_1 + x_0 + y_1) M_{11} (\bar{x}_1 + \bar{x}_0 + y_1) M_{14}$$

$$= M_1 (x_1 + \bar{y}_1) M_4 (\bar{x}_1 + y_1) M_{11} M_{14}$$

we can re-use M_3, M_6, M_9, M_{12} from rule 7a

$$M_1 M_3 = (x_1 + x_0 + y_1 + \bar{y}_0)(x_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (x_1 + x_0 + \bar{y}_0)$$
$$M_9 M_{11} = (\bar{x}_1 + x_0 + y_1 + \bar{y}_0)(\bar{x}_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (\bar{x}_1 + x_0 + \bar{y}_0)$$
$$M_4 M_6 = (x_1 + \bar{x}_0 + y_1 + y_0)(x_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (x_1 + \bar{x}_0 + y_0)$$
$$M_{12} M_{14} = (\bar{x}_1 + \bar{x}_0 + y_1 + y_0)(\bar{x}_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (\bar{x}_1 + \bar{x}_0 + y_0)$$

$$f = M_1 M_3 (x_1 + \bar{y}_1) M_4 M_6 (\bar{x}_1 + y_1) M_9 M_{11} M_{12} M_{14}$$

$$= (x_1 + x_0 + \bar{y}_0)(x_1 + \bar{y}_1)(x_1 + \bar{x}_0 + y_0)(\bar{x}_1 + y_1)(\bar{x}_1 + x_0 + \bar{y}_0)(\bar{x}_1 + \bar{x}_0 + y_0)$$

$$= (x_1 + \bar{y}_1)(\bar{x}_1 + y_1)(x_0 + \bar{y}_0)(\bar{x}_0 + y_0)$$

12) Implement the function from the truth table using NAND gates

m_i	x_1	x_2	x_3	f
m_0	0	0	0	0
m_1	0	0	1	1
m_2	0	1	0	1
m_3	0	1	1	0
m_4	1	0	0	1
m_5	1	0	1	0
m_6	1	1	0	0
m_7	1	1	1	1

$$f = \sum m(1,2,4,7) = m_1 + m_2 + m_4 + m_7$$

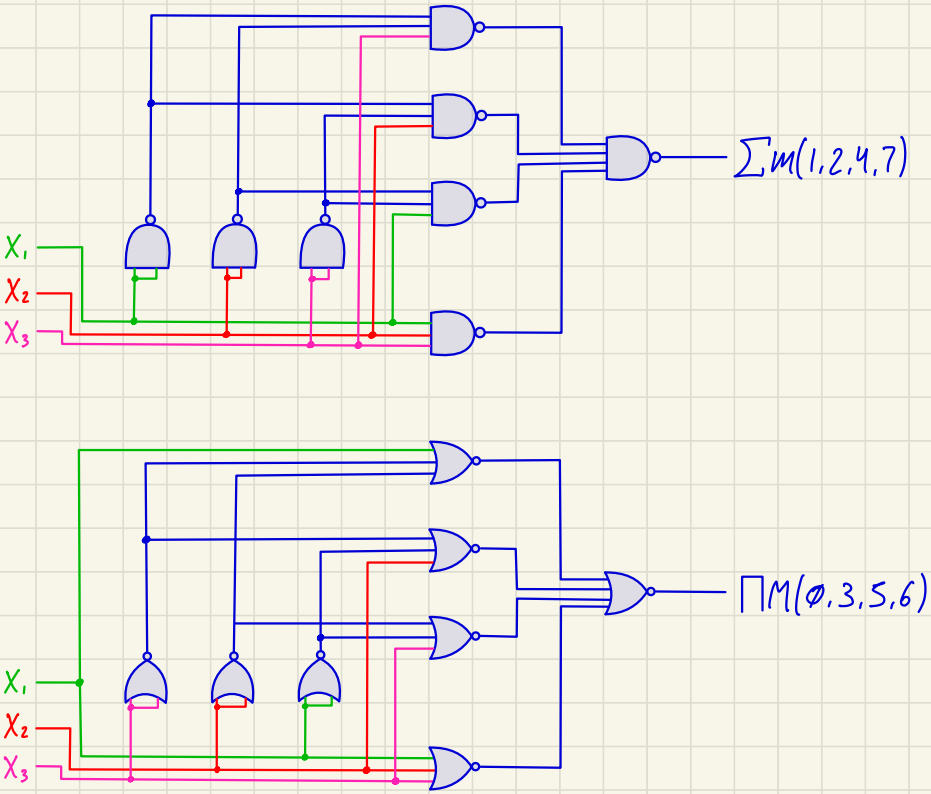
$$= \overline{x}_1\overline{x}_2x_3 + \overline{x}_1x_2\overline{x}_3 + x_1\overline{x}_2\overline{x}_3 + x_1x_2x_3$$

$$f = \prod M(0,3,5,6) = M_0M_3M_5M_6$$

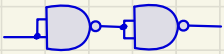
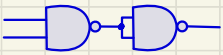
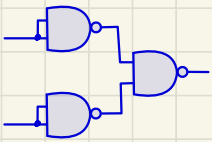
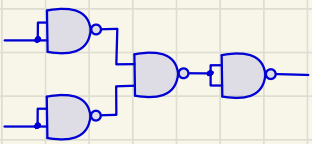
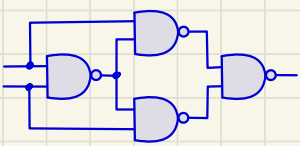
$$= (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + \overline{x}_3)(\overline{x}_1 + x_2 + \overline{x}_3)(\overline{x}_1 + \overline{x}_2 + x_3)$$

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

Diagrams



NAND Representation of the gates

NAND		
NOT		
BUFFER		
AND		
OR		
NOR		
XOR		
XNOR	